Language-Guided Adaptive Spatial–Frequency-Aware Network for Remote Sensing Change Detection

Chunpeng Wu, Shuli Cheng, *Member, IEEE*, Liejun Wang, and Yankang Liu

***Abstract*—Remote sensing change detection (RSCD) plays a crucial role in urban planning, disaster assessment, and ecological monitoring. However, the inherent complexity of remotely sensed images and the interference of nontargeted changes, such as illumination changes and ground renovation, often lead to the problem of misdetection of change areas by models. Moreover, many real-world changes occur at the semantic level rather than through explicit geometric or structural differences, making them difficult to distinguish based solely on visual appearance. Although deep learning-based methods have achieved significant progress in this field, most existing approaches struggle to effectively balance and address the complexities mentioned above due to insufficient semantic guidance. To this end, we propose a language-guided adaptive spatial–frequency-aware network (LGSFNet) to enhance the model's semantic guidance and discrimination ability for changing regions by introducing textual high-level semantic information at low cost. In this approach, textual prompts guide the model to focus more on change-related regions, thereby learning more discriminative features. Through disentangled fusion, this method provides a new perspective for traditional unimodal RSCD to utilize the high-level semantic information of text. Furthermore, we design a lightweight dual-domain collaborative sparse attention mechanism in both spatial and frequency domains, which mimics human vision by adaptively emphasizing salient information. Combined with wavelet transforms, this mechanism further enhances the model's sensitivity to complex change patterns, improving both discriminative power and robustness. Extensive experiments conducted on three widely used benchmarks—LEVIR, BCDD, and SYSU—demonstrate that LGSFNet achieves competitive performance compared with state-of-the-art methods.**

***Index Terms*—Binary change detection (BCD), contrastive language–image pretraining (CLIP), dual-domain collaborative sparse attention (DDCSA), language-guided, spatial–frequency-aware.**

I. Introduction

REMOTE sensing change detection (RSCD) identifies changes in surface objects or the environment over time by comparing remote sensing images captured at different periods. It has important applications in urban infrastructure planning [1], agricultural land monitoring [2], and disaster assessment [3]. However, remote sensing imagery (RSI) is inherently complex, and the challenges faced by RSCD can be broadly categorized into two representative issues. First, as shown in Fig. 1(a), nontarget variations (e.g., illumination changes, ground renovation) often introduce visually misleading changes between bi-temporal images, leading to false detections. Second, as shown in Fig. 1(b) and (c), many meaningful changes in real-world scenarios—such as vegetation degradation or road construction sites—are primarily reflected at the semantic level rather than through explicit geometric or structural differences. These subtle semantic changes are easily overshadowed by complex backgrounds when relying solely on visual features, resulting in missed detections. We argue that all of the above reflect the insufficient discriminative capability of existing methods. Effective RSCD requires not only sensitivity to changes at the appearance level, but also some semantic discrimination ability to distinguish meaningful changes from irrelevant visual changes in the model learning process. Therefore, designing an efficient RSCD network

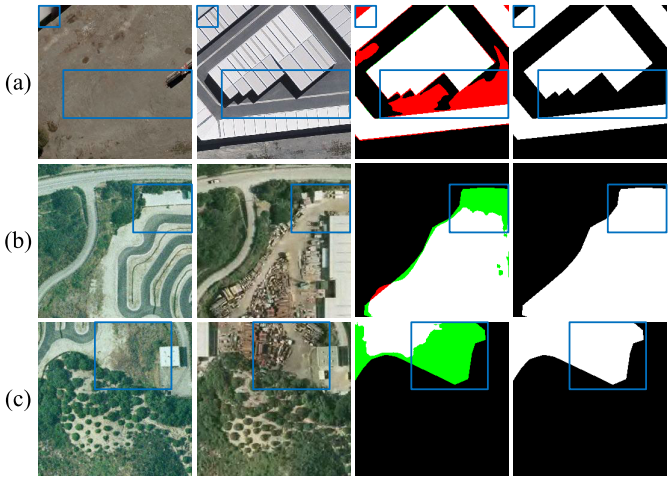


Fig. 1. Illustration of two representative challenges in RSCD. (a) Interference caused by nontarget variations leads to false detections. (b) and (c) Non-structural changes are difficult to distinguish using appearance-based features alone, resulting in missed detections. The first and second columns correspond to the bi-temporal images at T1 and T2, respectively. The third and fourth columns show the predicted change map and the ground truth. Changed regions are highlighted in white, while unchanged regions are shown in black. Red regions indicate false detections, and green regions denote missed detections.

Received 5 February 2026; revised 11 March 2026; accepted 17 April 2026. Date of publication 22 April 2026; date of current version 30 April 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2025YFF0515602 and in part by the National Natural Science Foundation of China under Grant 62562058. *(Corresponding authors: Shuli Cheng; Liejun Wang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3686099

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that can adapt to the complex characteristics of RSIs is crucial for improving detection performance and practical applicability.

Traditional change detection (CD) methods, such as principal component analysis (PCA) [4] and change vector analysis (CVA) [5], are structurally simple and offer high processing efficiency. However, these methods mainly rely on shallow pixel-level or spectral features, which limit their accuracy and robustness in complex scenarios. With the explosive growth of high-resolution RSI [6], [7], [8], deep learning-based methods have gained increasing attention. A large number of deep learning-based CD strategies [9], [10], [11] demonstrate that attention mechanisms [12], [13] and U-Net-like hierarchical fusion architectures [14], [15] play a significant role in improving detection accuracy. Among them, models such as convolutional neural networks (CNNs) [14], vision transformers (ViTs) [16], and Mamba [17] have become research hotspots due to their powerful feature representation capabilities. However, CNN-based attention mechanisms [18], [19] rely on local receptive fields. They struggle to capture long-range dependencies in complex change regions, limiting their performance in RSIs. ViT-based attention mechanisms [20], [21] possess global modeling capabilities, but their high computational complexity hinders efficiency and scalability in practical deployments. Recently, the emerging Mamba [22], [23] has demonstrated the ability to build global representations with linear complexity. However, similar to ViT, it still fails to effectively suppress large-scale static or redundant background information in RSIs, where weak signals from small local targets are easily overwhelmed [24]. To address this, many CD methods [25], [26], [27] attempt to combine CNN with ViT or Mamba to balance local and global modeling. Although these approaches achieve improved accuracy, they tend to be structurally redundant and computationally expensive. This makes them difficult to balance efficiency and practicality. More importantly, most existing methods rely solely on visual appearance, which fundamentally limits their ability to effectively distinguish semantic-level changes (e.g., vegetation degradation, road construction sites) from irrelevant visual variations in complex RSIs. Motivated by the above analysis, we argue that an effective RSCD framework should satisfy three key requirements: the ability to incorporate high-level semantic priors to distinguish meaningful changes from irrelevant visual variations; sufficient multiscale representation capacity to compensate for the limited expressiveness of lightweight visual backbones; and an adaptive attention mechanism that balances global semantic consistency with local structural sensitivity. To fulfill the requirement of semantic discrimination in RSCD, introducing external semantic priors becomes a natural and effective choice. In this context, visual language models, such as the contrastive language–image pretraining (CLIP) model [28], provide a promising approach for injecting high-level semantic information into CD models. However, existing multimodal CD approaches [29], [30] typically employ fully coupled image–text encoders, which significantly increase model parameters and computational cost, making them expensive in real deployment scenarios.

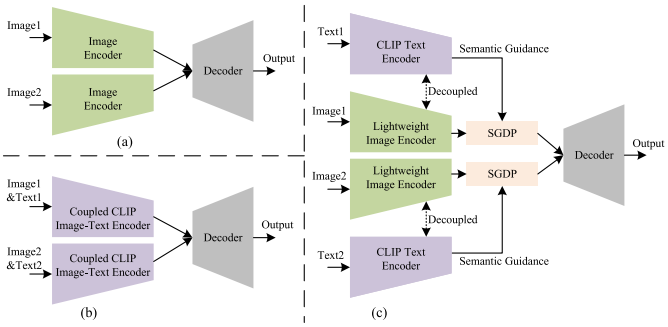


Fig. 2. Comparison of RSCD strategies. (a) Unimodal CD methods using a weight-shared image encoder. (b) Existing multimodal CD methods (e.g., ChangeCLIP) employ a coupled architecture, relying on computationally heavy encoders to process both images and text simultaneously for feature alignment. (c) LGSFNet (ours) adopts a decoupled strategy: it replaces the heavy visual backbone with a lightweight MobileNetV2 and utilizes the CLIP text encoder solely for semantic guidance via the SGDP module, significantly reducing Params. and FLOPs.

Following the above design principles, we propose a language-guided adaptive spatial–frequency-aware network (LGSFNet) for RSCD. Unlike existing methods (as shown in Fig. 2), LGSFNet does not adopt a fully coupled multimodal architecture. It uses a frozen CLIP text encoder as a semantic prior extraction module, and designs a semantic-guided differential perceiver (SGDP) to inject linguistic prior for image features in a decoupled and lightweight way. Meanwhile, a lightweight visual backbone is used for efficient feature extraction. To enhance the representation capacity of the lightweight backbone, a multiscale feature aggregation module (MSFA) is designed to strengthen multilevel structural and contextual features. Furthermore, a dual-domain collaborative sparse attention mechanism is introduced to jointly exploit spatial and frequency-domain information, enabling the model to further integrate semantic information and adaptively emphasize salient change patterns while suppressing background noise. Through the synergistic integration of semantic guidance, multiscale representation, and adaptive dual-domain attention, LGSFNet provides an effective and efficient solution to the inherent challenges of RSCD. The main contributions of this study are summarized as follows.

1) We propose a novel LGSFNet for RSCD, which introduces high-level semantic priors through an SGDP in a lightweight and decoupled manner, enabling the model to effectively distinguish meaningful semantic changes from irrelevant visual variations.
2) We design an MSFA, which integrates spatial features and semantic information across layers through progressive interactions, providing stronger semantic and structural support for lightweight backbones.
3) We develop a lightweight dual-domain collaborative sparse attention mechanism that jointly exploits spatial- and frequency-domain information to enhance semantic-aware change features, while suppressing redundant background interference.
4) Extensive experiments on three public datasets show that LGSFNet achieves a good balance between accuracy, robustness, and computational efficiency, realizing competitive performance.

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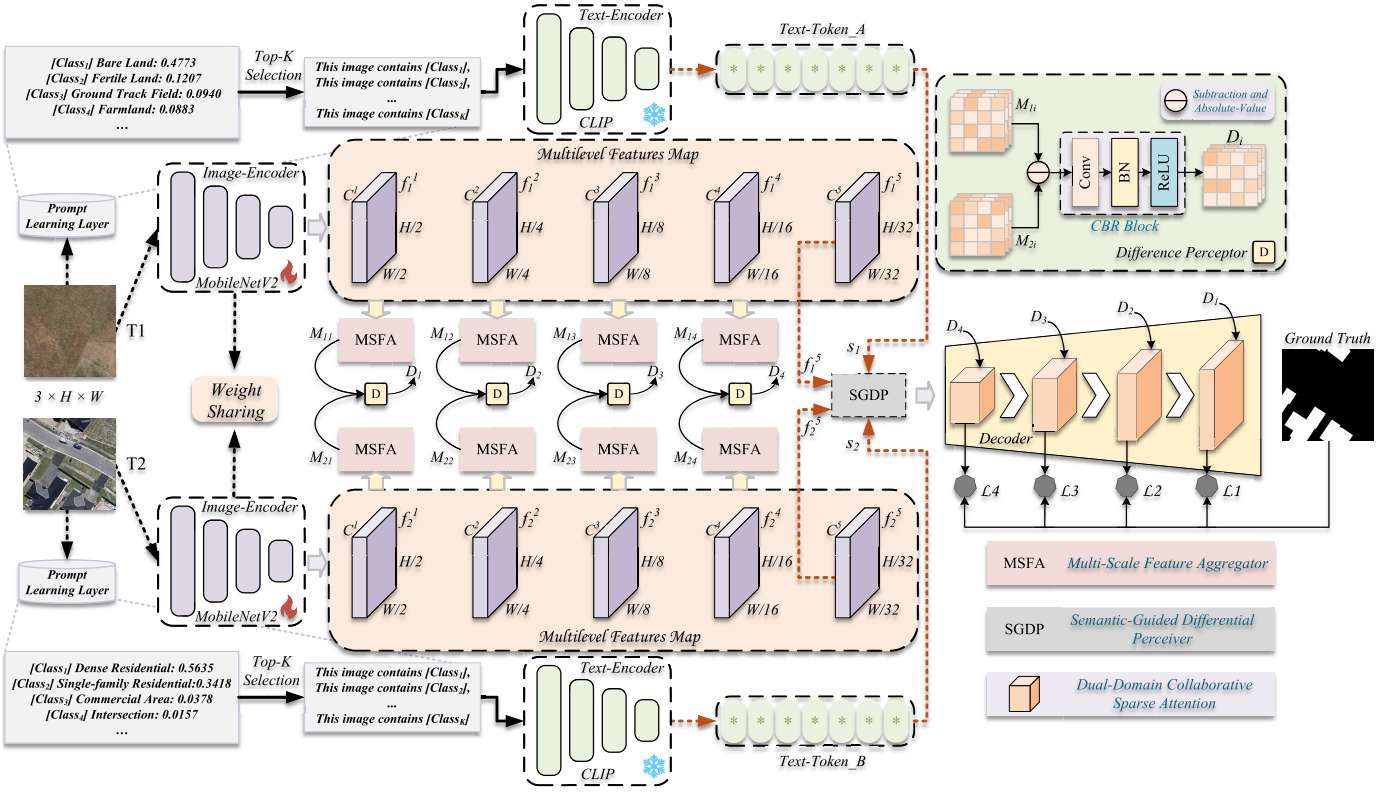


Fig. 3. Architecture of LGSFNet. Textual priors are first constructed via CLIP's zero-shot classification. A frozen CLIP text encoder is then used solely as a fixed semantic feature extractor to obtain textual representations, while MobileNetV2 extracts multilevel visual features from bi-temporal images. MSFA enhances multilevel image features, SGDP injects semantic guidance, and DDCSA captures multiscale change patterns to produce an accurate change map.

II. Related Work

In recent years, CD has made significant progress in the field of remote sensing. Researchers have found that effectively leveraging both local and global information plays a crucial role in improving detection accuracy, and have begun exploring architectural-level integration of CNNs with Transformers or Mamba. For example, MSTDSNet-CD [31] introduces a hybrid model based on a ResNet18 [32] backbone and multiscale Swin Transformer [33] to enhance the extracted feature maps. StyleFormer [25] proposes an iterative interaction strategy between Transformer and CNN features to alleviate the lack of local perception and nonhierarchical representations in ViT, thereby enhancing the model's representational capacity. Hybrid-MambaCD [24] proposed a Mamba-CNN hybrid architecture to obtain more representative global–local features with fewer model parameters. Although such hybrid architectures improve accuracy, they also introduce considerable architectural and computational redundancy. Notably, in CD tasks, there exists a significant foreground–background imbalance, where actual change regions occupy only a small portion of the image. Global modeling without distinction often introduces substantial background noise. Therefore, designing lightweight and efficient attention mechanisms that balance global awareness with local sensitivity remains a key research challenge in CD. To further enhance the model's ability to distinguish change features, ChangeCLIP [29] proposed a novel framework that leverages the strong semantic information from image–text pairs, marking the first exploration of vision–language models for CD. Building on this, VLCD [30] further explored the potential of textual information in conjunction with visual backbones for CD. However, these methods introduce both the visual and textual encoders of CLIP, which significantly increases model parameters and computational cost, making it expensive to incorporate textual information into unimodal CD tasks. Therefore, it is of great significance to explore a detachable vision–language CD framework with lower computational overhead.

III. Methods

A. Architecture Overview

The overall architecture of LGSFNet is shown in Fig. 3. We leverage the zero-shot image classification capability of the CLIP model to generate text prompts that match the semantics of bi-temporal remote sensing images. A predefined vocabulary of 56 remote sensing categories [29] is provided. For each image, CLIP selects the Top-K categories with the highest confidence scores and dynamically constructs K instance-specific textual descriptions (e.g., "this image contains [Class$_1$]", ..., "this image contains [Class$_K$]"). Compared with manually designed text templates, this strategy provides stronger adaptability and semantic consistency. Subsequently, the obtained bi-temporal text prompts are input into a frozen, pretrained CLIP text encoder, which is used solely as a fixed semantic feature extractor to obtain high-level semantic representations, denoted as $S = \{s_1, s_2\}$, where s_1 and s_2 represent the text tokens corresponding to temporal phases

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Time 1 (T1) and Time 2 (T2), respectively. Meanwhile, T1 and T2 are input into a shared-weight pretrained MobileNetV2 [34] image encoder to extract multilevel image representations, denoted as $f_i = \{f_i^1, f_i^2, f_i^3, f_i^4, f_i^5\}, i \in \{1, 2\}$. Then, we apply MSFA to each level of the encoded features to enhance the representation capability of the lightweight backbone, formulated as

$$M_{ij} = \mathrm{MSFA}_j(f_i), \quad i \in \{1, 2\}, \ j \in \{1, 2, 3, 4\} \tag{1}$$

where M_{ij} denotes the enhanced features of the jth layer from Ti. To reduce computational cost, we obtain the difference features in a simple way

$$D_j = \mathrm{C_1BR}\left(|M_{1j} - M_{2j}|\right), \quad j \in \{1, 2, 3, 4\} \tag{2}$$

where $|\cdot|$ denotes element-wise subtraction followed by absolute operation, and $\mathrm{C_1BR}(\cdot)$ denotes a 1×1 convolution layer with batch normalization and activation. Meanwhile, we adopt SGDP to integrate the text features S with the deep image features f_1^5 and f_2^5 to achieve effective vision–language feature collaboration

$$D_s = \mathrm{SGDP}\left(f_1^5, f_2^5, s_1, s_2\right) \tag{3}$$

where D_s denotes the difference features enhanced by textual information. Next, we introduce dual-domain collaborative sparse attention (DDCSA) to capture more discriminative change information while suppressing extraneous noise in shallow features. This process is formulated as

$$F_4 = \mathrm{DDCSA}\left(\mathrm{Up}(D_s), D_4\right) \tag{4}$$

$$F_j = \mathrm{DDCSA}\left(\mathrm{Up}\left(F_{j+1}\right), D_j\right), \quad j \in \{3, 2, 1\} \tag{5}$$

where F_4 denotes the initial decoding features from the first decoder stage. F_j represents the decoding features at each respective layer, while F_{j+1} refers to the features passed from the preceding (deeper) stage. Up$(\cdot)$ denotes bilinear interpolation. Finally, a clear change map is obtained. To mitigate the sample imbalance issue during training, we employ a deep supervision strategy [35] and utilize the hybrid loss from A2Net [36] to constrain the model.

B. Multiscale Feature Aggregation

In RSCD tasks, the feature extraction capability of the backbone network largely determines the model's ability to identify fine-grained changes. To boost a lightweight backbone's representation capacity, we propose MSFA. It builds dense feature interaction paths between the backbone's encoding layers. This enhances multilevel structural information modeling.

As shown in Fig. 4, MSFA takes the current layer feature as the main input and combines it with auxiliary features from adjacent layers to construct a main-auxiliary collaborative fusion structure that integrates multiscale information. Taking f_2^3 as an example, multilevel features from f_2 are fed into MSFA, where f_2^3 serves as the main feature and the others as auxiliary. The main feature is processed by a 3×3 convolution with batch normalization and ReLU to enhance representation and suppress noise. Auxiliary features

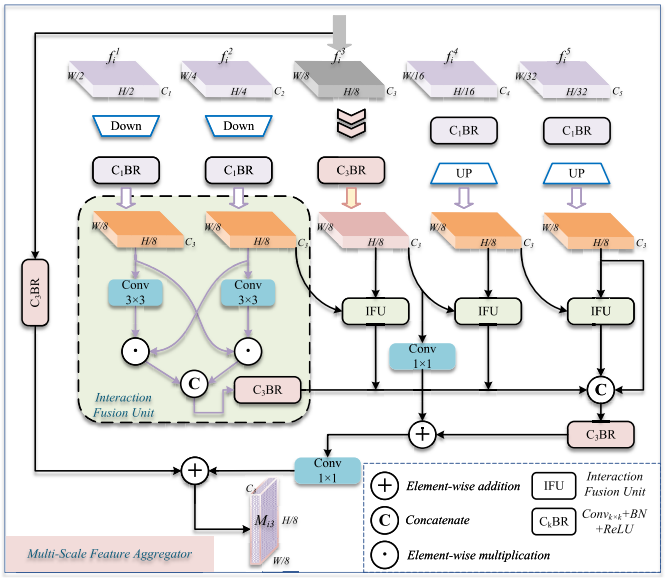


Fig. 4. Multiscale feature aggregation.

are spatially aligned (via pooling or upsampling) and channel-adjusted (via 1×1 convolution) to unify the scale. This process can be expressed as

$$F_2^{(j,3)} = \begin{cases} \mathrm{C_1BR}\left(\mathrm{MP}\left(f_2^j\right)\right), & j < 3 \\ \mathrm{C_3BR}\left(f_2^j\right), & j = 3 \\ \mathrm{Up}\left(\mathrm{C_1BR}\left(f_2^j\right)\right), & j > 3 \end{cases} \tag{6}$$

where MP$(\cdot)$ denotes max pooling, Up$(\cdot)$ denotes bilinear interpolation, and $\mathrm{C}_k\mathrm{BR}(\cdot)$ denotes a $k \times k$ convolution layer with batch normalization and ReLU. $F_2^{(j,3)}$ represents the spatially and channel-aligned feature from f_2^j. Next, we design a lightweight interaction fusion unit (IFU) to build bidirectional interaction paths between adjacent layers, enabling complementary enhancement between features. This process can be expressed as

$$F_2^3 = \mathrm{C_3BR}\left(\mathrm{Cat}\left(\mathrm{IFU}^{(1,2)}, \mathrm{IFU}^{(2,3)}, \mathrm{IFU}^{(3,4)}\right.\right. \\ \left.\left.\mathrm{IFU}^{(4,5)}, F_2^{(5,3)}\right)\right) \tag{7}$$

$$\mathrm{IFU}^{(j,j+1)} = \mathrm{C_3BR}\left(\mathrm{Cat}\left(\mathrm{Conv}_{3\times3}\left(F_2^{(j,3)}\right) \cdot F_2^{(j+1,3)}\right.\right. \\ \left.\left.\mathrm{Conv}_{3\times3}\left(F_2^{(j+1,3)}\right) \cdot F_2^{(j,3)}\right)\right) \tag{8}$$

where $\mathrm{IFU}^{(j,j+1)}$ denotes the interaction between adjacent layers, and Cat$(\cdot)$ denotes feature concatenation. A residual connection is applied to preserve the integrity of the main feature path. This process is formulated as

$$M_{23} = \mathrm{Conv}_{1\times1}\left(F_2^3 + \mathrm{Conv}_{1\times1}\left(F_2^{(3,3)}\right)\right) + \mathrm{C_3BR}\left(f_2^3\right) \tag{9}$$

where M_{23} denotes the enhanced feature at the third level of temporal phase T2. The introduction of MSFA provides a more robust multiscale feature foundation for subsequent decoding stages.

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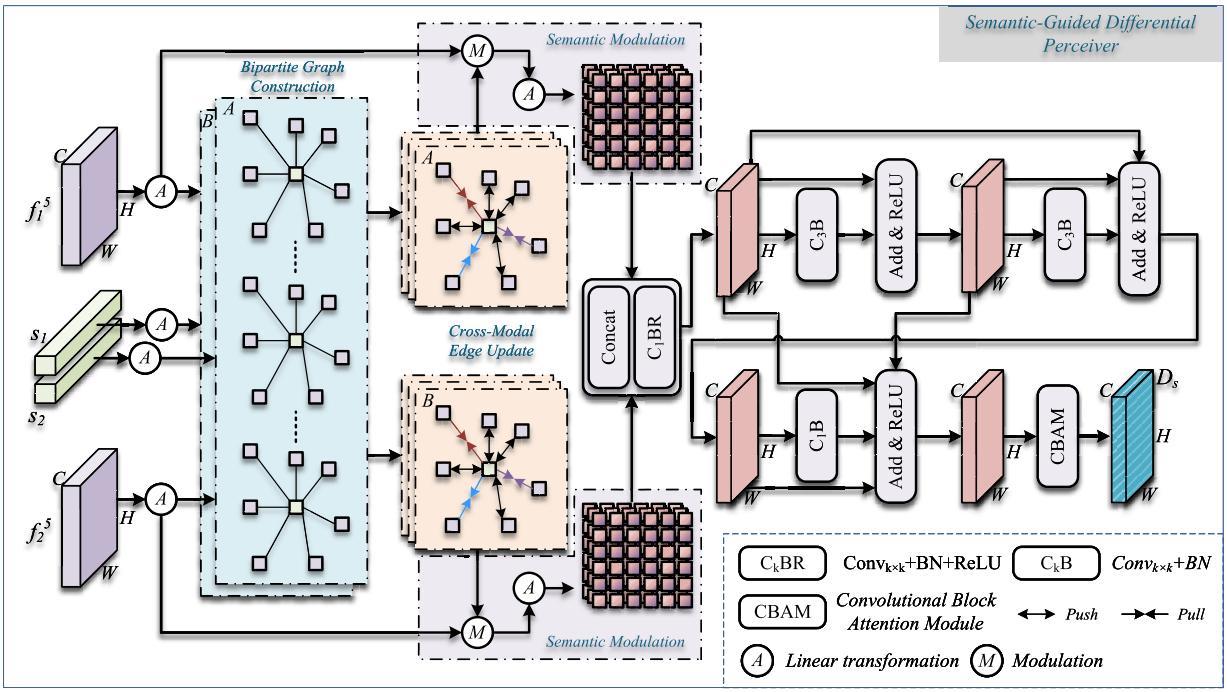


Fig. 5. Semantic-guided differential perceiver.

C. Semantic-Guided Differential Perceiver

In graph theory, a bipartite graph $G = (L, P, E)$ consists of two disjoint sets of nodes, where all edges E connect a node in L to a node in P. This topology is naturally suitable for modeling the interactions between two sets of heterogeneous entities, providing an effective path for cross-modal information dissemination and feature aggregation. To effectively bridge textual semantics and visual representations, we propose SGDP, as shown in Fig. 5. This module constructs multiple star bipartite graphs between K dynamic text nodes and image features for efficient cross-modal semantic guidance.

Taking time phase T1 as an example, the visual feature $f_1^5 \in \mathbb{R}^{C\times H\times W}$ is reshaped into a sequence $f_1^5 \in \mathbb{R}^{N\times C}$ (where $N = H \times W$), while the textual feature $s_1 \in \mathbb{R}^{K\times D}$ serves as the source for semantic anchors. We first project the visual feature f_1^5 and the textual feature s_1 into a unified d-dimensional latent space to instantiate the node sets. The visual node set P and the textual node set L are defined as

$$P = \left\{\mathbf{x}_i \mid \mathbf{x}_i = \text{Linear}_P\left(f_{1,i}^5\right)\right\}_{i=1}^{N} \tag{10}$$

$$L = \left\{\mathbf{t}_j \mid \mathbf{t}_j = \text{Linear}_L\left(s_{1,j}\right)\right\}_{j=1}^{K} \tag{11}$$

where $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{t}_j \in \mathbb{R}^d$ represent the d-dimensional vector nodes corresponding to the ith visual token and the jth semantic anchor, respectively. K and N represent the number of textual anchors and visual tokens, respectively. Subsequently, the edge set E is defined as the set of all potential connections between text nodes and visual nodes. In order to accurately model this cross-modal interaction, we define the weights of edges $e_{i,j} \in E$ as the attentional choices of the visual markers for specific semantic anchor points. Specifically, for each visual node $\mathbf{x}_i$, the strength of its association with all text nodes $\mathbf{t}_j$ is initially measured by dot product similarity

$$M_{j,i} = \frac{\mathbf{t}_j \mathbf{x}_i^T}{\sqrt{d}}, \quad \mathbf{M} \in \mathbb{R}^{K\times N} \tag{12}$$

where $\mathbf{M}$ is the similarity matrix. d is the scaling factor. In this process, the textual nodes serve as guidance signals, establishing heterogeneous connection weights across different visual regions. By imposing a Softmax normalization constraint along the textual dimension, each visual token undergoes an "information reception competition" based on its semantic fit with different textual anchors. This weight assignment process is represented as

$$w_{i,j} = \frac{\exp\left(M_{j,i}\right)}{\sum_{k=1}^{K}\exp\left(M_{k,i}\right)}, \quad \mathbf{W} \in \mathbb{R}^{N\times K} \tag{13}$$

where $w_{i,j}$ denotes the dynamic edge weight of the bipartite graph. Here, the matrix $\mathbf{W}$ is the transpose-normalized form of $\mathbf{M}$, where each row represents the attention distribution of a visual token over all K textual nodes. Based on these weights, the semantic information of the text nodes is selectively aggregated and injected into the corresponding visual nodes to complete the modulation and enhancement of the visual features

$$\hat{\mathbf{x}}_i = \mathbf{x}_i + \sum_{j=1}^{K} w_{i,j}\mathbf{t}_j \tag{14}$$

where the summation term realizes the weighted aggregation from the textual domain to the visual domain. Finally, to reintegrate the enhanced nodes back into the visual representation, the set of modulated nodes $\hat{P} = \{\hat{\mathbf{x}}_i\}_{i=1}^{N}$ is mapped back to the original feature dimension and spatially reshaped to form the output feature map $\hat{f}_1^5 \in \mathbb{R}^{C\times H\times W}$. This operation

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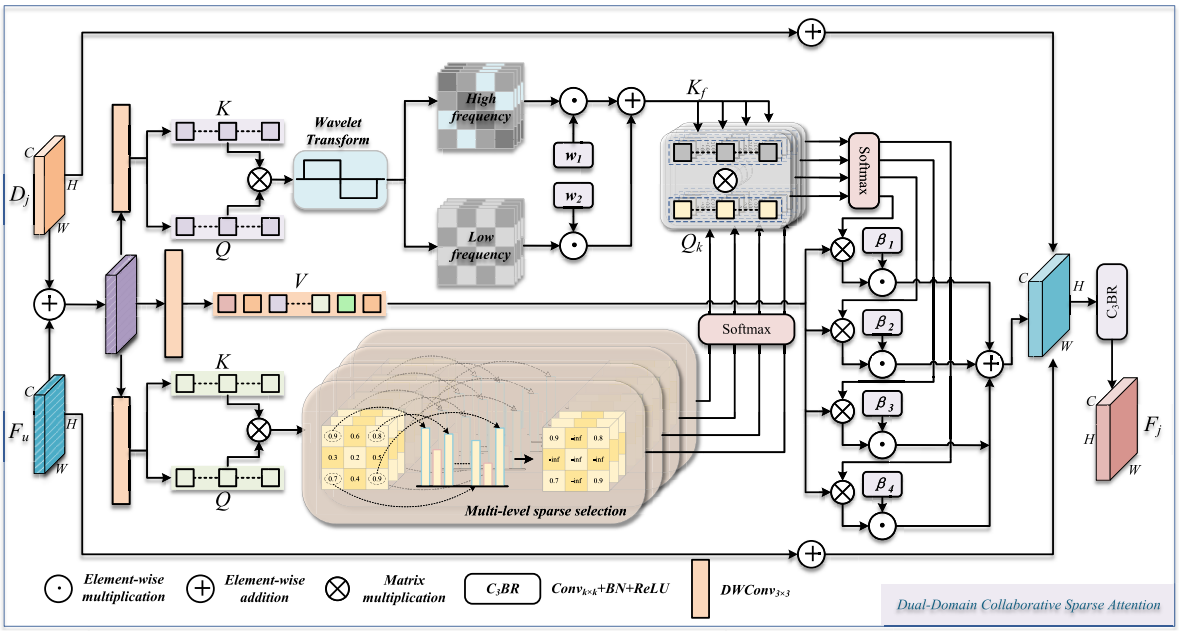


Fig. 6. Dual-domain collaborative sparse attention.

effectively restores the spatial topology of the visual tokens while preserving the newly incorporated semantic guidance.

Similarly, we obtain the modulated feature map $\hat{f}_2^5 \in \mathbb{R}^{C\times H\times W}$ for time phase T2 using the same mechanism. By strengthening the semantic information within the bi-temporal features, respectively, the module facilitates more discriminative change modeling. Next, we perform channel concatenation on the enhanced feature maps. We then employ a series of convolutional blocks with dense residual connections to effectively model the semantic differences. The process is formulated as follows:

$$\begin{aligned}\mathbf{r}_1 &= \mathrm{C_1BR}\left(\mathrm{Cat}\left(\hat{f}_1^5, \hat{f}_2^5\right)\right)\\ \mathbf{r}_2 &= \mathrm{ReLU}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{r}_1\right) + \mathbf{r}_1\right)\\ \mathbf{r}_3 &= \mathrm{ReLU}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{r}_2\right) + \mathbf{r}_2 + \mathbf{r}_1\right)\\ F_{diff} &= \mathrm{ReLU}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{r}_3\right) + \mathbf{r}_3 + \mathbf{r}_2 + \mathbf{r}_1\right).\end{aligned} \tag{15}$$

Subsequently, the difference feature map is further refined using the convolutional block attention module (CBAM) [12] to obtain the final variation map D_s

$$D_s = \mathrm{O}\left(F_{diff}\right) \tag{16}$$

where $\mathrm{O}(\cdot)$ denotes a CBAM operation. It adaptively recalibrates change-sensitive regions through joint channel and spatial attention mechanisms. This hierarchical refinement process ensures that the model attends to the most significant semantic transitions between bi-temporal images.

D. Dual-Domain Collaborative Sparse Attention

To further enhance the representation capability of semantic-enhanced features in change localization and reconstruction, we propose DDCSA. Without introducing extra structural redundancy or computational overhead, DDCSA efficiently models the synergy between structural details and semantic dependencies.

As shown in Fig. 6, DDCSA integrates structural parsing in the frequency domain with sparse cross-channel selection in the spatial domain, forming a complementary and collaborative attention mechanism. Specifically, the input to the jth DDCSA module is twofold: the MSFA-refined difference features $D_j \in \mathbb{R}^{C\times H\times W}$ from the symmetrical encoder stage, and the spatially aligned semantic features $F_u = \mathrm{Up}(F_{j+1}) \in \mathbb{R}^{C\times H\times W}$ derived from the previous decoder level. Element-wise addition is first applied to fuse the inputs. The result is then processed by depth-wise separable convolution (DWConv) to extract the initial attention input

$$Q, K, V = \mathrm{Reshape}\left(\mathrm{Chunk}\left(\mathrm{DWConv}\left(D_j + F_u\right)\right)\right) \tag{17}$$

where $\mathrm{Chunk}(\cdot)$ denotes equal partitioning along the channel dimension, and $\mathrm{Reshape}(\cdot)$ rearranges the tensor shape. Q, K, and V are reshaped to $\mathbb{R}^{d\times HW}$, with $d = C/4$ as each attention head's dimension. The dense channel attention matrix $A \in \mathbb{R}^{d\times d}$ is obtained via matrix multiplication of Q and the transpose of K. Compared to conventional spatial attention, this approach significantly reduces computational cost while preserving global semantic dependencies across channels [37]. On this basis, we apply the top-k strategy [38] to construct multiple sparse query subsets $Q_k \in \mathbb{R}^{d\times d}$ from A

$$Q_k = \left[\mathrm{Top}_k\left(A\right)\right]_{ij} = \begin{cases} A_{ij}, & A_{ij} \geq \mathrm{top}\left(k\right) \\ -\infty, & A_{ij} < \mathrm{top}\left(k\right) \end{cases} \tag{18}$$

where $\mathrm{top}(k)$ denotes the kth largest value in the jth row of A, and $\mathrm{Top}_k(\cdot)$ denotes top-k selection operator. By introducing different k values, the design enables the model to capture channel interactions at different sparsity levels. Specifically, smaller k values focus on the most salient semantic associations, while larger k values incorporate a wider and more diverse range of channel dependencies, thus enhancing the robustness of feature selection. Meanwhile, a two-level wavelet packet decomposition (WPD) is applied to decompose the attention matrix A into high-frequency and low-frequency

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components, denoted as A_{HH} and A_{LL}. Physically, A_{LL} represents stable and consistent semantic dependencies between channels. It captures the co-occurrence patterns shared by large-scale features and ensures the overall continuity of the predicted change map. In contrast, A_{HH} captures sharp shifts in channel interactions that are physically associated with structural boundaries, fine-grained textures, and localized changes

$$A_{LL_{pre}}, A_{HH_{pre}} = \mathrm{DWT}(A) \tag{19}$$

$$A_{LL} = \mathrm{DWT}\left(A_{LL_{pre}}\right) \tag{20}$$

$$A_{HH} = \mathrm{DWT}\left(A_{HH_{pre}}\right) \tag{21}$$

where DWT(·) denotes a single 2-D Haar Discrete Wavelet Transform. A_{HH} suppresses redundant correlation with other irrelevant channels (e.g., background noise channels) via ReLU. A_{LL} enhances the overall consistency by Softmax. These two branches are then merged to obtain the frequency modulation kernel $K_f \in \mathbb{R}^{d\times d}$

$$K_f = w_1 \cdot \mathrm{ReLU}(A_{HH}) + w_2 \cdot \mathrm{Softmax}(A_{LL}) \tag{22}$$

where w_1 and w_2 are learnable weights that balance detail sensitivity and semantic integrity. Next, each Q_k is modulated by K_f and multiplied with V to generate the representation $R_k \in \mathbb{R}^{HW\times d}$. This process accomplishes a deep synergy between Q_k (responsible for selecting key channel combinations) and K_f (responsible for correcting the interaction quality). Meanwhile, feature reconstruction for structure perception is realized by weighting the spatial location based on multiscale channel affinity. This process is formulated as

$$R_k = \mathrm{Softmax}\left(\mathrm{Softmax}(Q_k) \otimes K_f^{\top}\right) \otimes V \tag{23}$$

where ⊗ denotes matrix multiplication. We dynamically fuse these feature representations by leveraging learnable weights to enhance structural restoration and scale adaptability. In addition, a residual connection is introduced to stabilize the feature flow. This process is formulated as

$$F_j = \mathrm{C_3BR}\left(F_u + D_j + \mathrm{Reshape}\left(\sum_{i=1}^{n} \beta_i \cdot R_k^i\right)\right) \tag{24}$$

where Reshape(·) rearranges the tensor shape. β_i is learnable weights, R_k^i denotes the ith attention feature representation, and n is the number of such representations. This mechanism significantly improves the model's ability to spatially reconstruct complex scenarios while enhancing the discriminative power of change information.

E. Loss Function

CD tasks are typically characterized by a severe imbalance between changed and unchanged pixels, where the unchanged regions dominate most scenes. This imbalance can negatively affect the optimization process and lead to biased predictions toward the majority class. To address this issue, we adopt a hybrid loss function that combines binary cross-entropy (BCE) loss and Dice loss, enabling both pixel-level supervision and region-level consistency. The BCE loss is formulated as

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[g_i \log(\hat{p}_i) + (1-g_i)\log(1-\hat{p}_i)\right] \tag{25}$$

where N denotes the total number of pixels, $g_i \in \{0,1\}$ represents the ground-truth label of the ith pixel, and $\hat{p}_i \in [0,1]$ indicates the predicted probability of change. This term enforces accurate pixel-wise classification during training. To further mitigate the influence of class imbalance and emphasize the spatial agreement between predictions and ground truth, the Dice loss is employed, which is defined as

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_{i=1}^{N} g_i \hat{p}_i}{\sum_{i=1}^{N} g_i^2 + \sum_{i=1}^{N} \hat{p}_i^2}. \tag{26}$$

Compared with BCE loss, the Dice loss focuses more on region-level overlap and is less sensitive to the proportion of changed pixels. The final training objective is obtained by combining the two loss terms

$$\mathcal{L}_{\mathrm{total}} = \sum_{k=1}^{n}\left(\mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{Dice}}\right) \tag{27}$$

where n denotes the number of prediction outputs involved in the loss computation. By jointly optimizing BCE and Dice losses, the proposed objective balances pixel-level discrimination and global structural consistency, thereby improving robustness under severe class imbalance.

IV. Experiments and Results Analysis

This section first outlines the evaluation metrics and datasets used in our study. We then report quantitative and qualitative results on three widely used benchmarks. Furthermore, ablation studies are conducted to validate the effectiveness of the proposed method.

A. Datasets and Evaluation Metrics

To evaluate the practical performance of LGSFNet, we conducted experiments on three widely used public datasets: LEVIR [39], BCDD [40], and SYSU [41]. They cover various change scenarios, including buildings, ships, roads, and vegetation. Following the cropping strategy commonly adopted in existing works, all image samples are resized to a uniform size of 256 × 256. Detailed information of each dataset is summarized in Table I. We adopt precision (Pre.), recall (Rec.), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA) as evaluation metrics, where higher values indicate better performance. Their mathematical definitions are given as follows:

$$Pre. = \frac{TP}{TP+FP} \tag{28}$$

$$Rec. = \frac{TP}{TP+FN} \tag{29}$$

$$F1 = \frac{2\times Pre.\times Rec.}{Pre.+Rec.} \tag{30}$$

$$IoU = \frac{TP}{TP+FP+FN} \tag{31}$$

$$OA = \frac{TP+TN}{TP+TN+FP+FN} \tag{32}$$

where TP denotes the number of changed pixels that are correctly predicted as changed. TN denotes the number of unchanged pixels that are correctly predicted as no-change.

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TABLE I
DETAILED INFORMATION ON THE THREE DATASETS

Datasets	Type	Spatial Resolution	Size/image	Number of Pixels			Number of Dataset		
				Changed	Unchanged	Ratio	Train	Val	Test
LEVIR	Building	0.5 m/pixel	256×256	30,913,975	637,028,937	1:20.61	7120	1024	2048
BCDD	Building	0.2 m/pixel	256×256	21,352,815	477,759,663	1:22.37	5895	794	745
SYSU	Complex	0.5 m/pixel	256×256	286,092,024	1,024,627,976	1:3.58	12000	4000	4000

FP denotes the number of unchanged pixels that are incorrectly predicted as changed. FN denotes the number of changed pixels that are incorrectly predicted as no-change. *Pre*. measures the reliability of the predictions. *Rec*. evaluates the model's ability to detect changed regions. $F1$ provides a harmonic balance between Precision and Recall. *IoU* quantifies the consistency between the predicted regions and the ground truth. *OA* reflects the overall classification accuracy. In addition, model parameters (Params.) and computational cost (FLOPs) are also reported to assess efficiency.

B. Implementation Details

To ensure fairness and reproducibility, all experiments were conducted on an NVIDIA A40 GPU with 48-GB memory. Standard data augmentation techniques, including horizontal and vertical flipping, resizing, and random cropping, were applied to enhance the robustness of model training. We employed the Adam optimizer with a momentum of 0.9 and a weight decay of $1e^{-4}$. The initial learning rate was set to 0.001 with a batch size of 32. During training, a polynomial learning rate schedule was used to gradually decay the learning rate to zero over the course of 40 000 iterations. For the SYSU dataset, training was conducted for 25 000 iterations. The model parameters corresponding to the highest validation $F1$-score were saved for testing to ensure optimal performance.

C. Comparison With Advanced Method

To evaluate the performance of LGSFNet, we conducted comparative experiments against nine state-of-the-art methods on three public benchmark datasets with distinct characteristics. These include ChangeFormer [21], DMINet [42], STENet [9], ChangeCLIP [29], CF-GCN [43], ELGC-Net [44], ChangeRD [45], STRobustNet [46], and CDMamba [11]. These methods are briefly introduced below.

1) *ChangeFormer:* This method leverages a Transformer and multilayer perceptron (MLP) to capture global relationships and efficiently handle the RSCD task.
2) *DMINet:* This method suppresses interference and focuses on regions of change by integrating self-attention and cross-attention mechanisms. It further enhances detection performance through a multilevel aggregation strategy.
3) *STENet:* This method employs a dynamic selective attention mechanism and 3-D convolutions to construct temporal and spatial dual pathways, effectively capturing changing features.
4) *ChangeCLIP:* This method proposes a multimodal CD approach that enhances the model's understanding of changes by incorporating textual information.
5) *CF-GCN:* This method captures long-range dependencies and feature interactions in dual-time images through coordinate-space convolutions, while suppressing interference from "nonsemantic variations" via boundary-aware and knowledge recall modules.
6) *ELGC-Net:* This method combines local context-aware deep convolutions with global context pooling and transposed attention mechanisms to achieve efficient multiscale feature aggregation.
7) *ChangeRD:* This method mitigates the impact of registration accuracy on CD by spatially transforming intermediate-layer features in the Siamese network, and alleviates spectral inconsistency via the AgCDC module.
8) *STRobustNet:* This method generates spatiotemporal robust representations for land use categories via the RRGModule, transforming dual-temporal features into consistent categorical representations to mitigate spatiotemporal inconsistencies.
9) *CDMamba:* This method combines Mamba's global feature extraction capabilities with CNN's strengths in local feature extraction, thereby enabling the capture of more comprehensive image representations.

The following presents the quantitative and qualitative analysis of the comparative experiment.

Quantitative Evaluation: To ensure fairness and representativeness in evaluation, we employed two widely used CLIP pretrained weights to assess LGSFNet. As shown in Tables II and III, the proposed model achieves competitive performance across all three benchmark datasets, which preliminarily validates the effectiveness of its overall framework. Compared with unimodal approaches, LGSFNet (RN50) surpasses STENet on the LEVIR by 0.91% in $F1$ and 1.53% in IoU, and outperforms CDMamba on the BCDD by 1.93% in $F1$ and 3.45% in IoU. On the SYSU, our method achieves an improvement of 2.24% in $F1$ and 3.27% in IoU over STENet. Furthermore, compared with ChangeCLIP, a model that also incorporates language information, our method significantly reduces model complexity while maintaining comparable inference speed (approximately 22 FPS) and competitive detection performance. Specifically, the number of parameters is reduced from 63.82 to 29.73 M, representing a decrease of approximately 53%, and the FLOPs are reduced from 35.55 to 10.56 G, corresponding to a reduction of about 70%. These results indicate that, without sacrificing inference efficiency,

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TABLE II
QUANTITATIVE EXPERIMENTAL RESULTS ON LEVIR AND BCDD. THE BEST RESULTS ARE SHOWN IN BOLD. THE UNDERLINE REPRESENTS THE BEST RESULT OF THE UNIMODAL ARCHITECTURE

Model	FLOPs (G)	Params. (M)	FPS	LEVIR					BCDD				
				F1 (%)	IoU (%)	OA (%)	Rec. (%)	Pre. (%)	F1 (%)	IoU (%)	OA (%)	Rec. (%)	Pre. (%)
ChangeFormer	202.79	41.03	17.61	90.20	82.16	99.02	88.47	92.01	90.05	81.90	99.10	87.28	93.00
DMINet	14.55	6.24	175.79	90.74	83.05	99.07	89.25	92.28	87.02	77.02	98.77	88.12	85.94
STENet	14.72	10.95	60.30	90.86	83.26	99.09	89.35	92.43	91.53	84.38	99.22	90.34	92.75
ChangeCLIP (RN50)	35.55	63.82	22.52	91.52	84.37	99.15	90.21	**92.88**	94.61	89.77	99.50	93.26	96.00
CF-GCN	44.08	13.58	54.36	90.51	82.67	99.04	89.53	91.51	88.50	79.37	98.99	83.29	94.40
ELGC-Net	187.99	10.57	18.40	89.95	81.73	98.99	88.52	91.42	89.12	80.38	99.03	85.03	93.64
ChangeRD	18.27	29.47	41.28	89.63	81.20	98.97	87.80	91.53	90.09	81.97	99.11	86.32	94.20
STRobustNet	13.19	5.23	13.82	90.86	83.26	99.08	90.19	91.54	90.96	83.42	99.19	**95.05**	87.21
CDMamba	49.26	11.91	3.96	90.66	82.91	99.06	89.76	91.58	93.11	87.10	99.37	90.54	95.83
Ours (ViT-B/32)	10.56	29.73	22.83	91.63	84.56	99.16	90.47	92.82	94.69	89.91	99.51	92.89	**96.55**
Ours (RN50)	10.56	29.73	22.80	**91.77**	**84.79**	**99.17**	**91.13**	92.42	**95.04**	**90.55**	**99.54**	93.73	96.40

TABLE III
QUANTITATIVE EXPERIMENTAL RESULTS ON SYSU. THE BEST RESULTS ARE SHOWN IN BOLD. THE UNDERLINE REPRESENTS THE BEST RESULT OF THE UNIMODAL ARCHITECTURE

Model	SYSU				
	F1 (%)	IoU (%)	OA (%)	Rec. (%)	Pre. (%)
ChangeFormer	78.79	65.01	90.33	76.14	81.63
DMINet	81.14	68.26	91.56	76.92	85.84
STENet	81.55	68.84	91.36	80.97	82.14
ChangeCLIP (RN50)	83.62	71.85	**92.49**	81.28	**86.10**
CF-GCN	77.96	63.88	90.50	71.23	86.10
ELGC-Net	78.82	65.05	90.52	74.81	83.28
ChangeRD	76.89	62.45	89.56	73.64	80.43
STRobustNet	79.03	65.32	90.64	74.75	83.82
CDMamba	79.17	65.53	90.61	75.70	82.98
Ours (ViT-B/32)	83.62	71.84	92.34	82.93	84.31
Ours (RN50)	**83.79**	**72.11**	92.38	**83.52**	84.07

our method can effectively exploit textual information using a more compact model architecture. This thereby avoids the computational redundancy caused by directly employing large multimodal encoders.

Qualitative Evaluation: The visual comparisons presented in Fig. 7 provide intuitive evidence that complements and reinforces the quantitative analysis. In the figure, white regions denote correctly detected change areas, black regions indicate correctly identified unchanged areas, red regions correspond to false positives, and green regions represent false negatives. On the SYSU, the scenes contain numerous irregular change targets. As illustrated in Fig. 7(a)–(e), all comparison methods suffer from varying degrees of omission (green regions), failing to fully extract all true change areas. In contrast, the prediction maps generated by our LGSFNet exhibit the smallest green regions, indicating its stronger sensitivity to weak change signals under complex backgrounds and its superior capability to reduce missed detections. On the LEVIR [Fig. 7(f)–(j)], building changes are often interfered with by irrelevant signals such as trees and illumination variations. These factors tend to obscure real target changes, leading to extensive missed detections (green regions) in competing methods. Although our method still has room for improvement in certain boundary details, the total area of erroneous pixels (red and green regions) in its predictions is significantly smaller than that of other methods. This demonstrates the effectiveness of incorporating language priors and the dual-domain attention mechanism in suppressing irrelevant changes and enhancing feature discriminability. On the higher resolution BCDD [Fig. 7(k)–(o)], irrelevant objects become more prominent and visually evident. For nonbuilding changes (e.g., shadows or ground textures), the prediction results of comparison methods contain noticeable noise and pseudo changes. Our method exhibits remarkable robustness against such interference, producing cleaner and more complete change maps with sharper boundaries. These results further validate the strong robustness of LGSFNet in handling the inherent complexity of high-resolution imagery.

In summary, across all visualized results, LGSFNet achieves the best balance between reducing false positives (red) and minimizing false negatives (green). This observation is fully consistent with the conclusions drawn from Tables II and III, where LGSFNet attains the highest $F1$—the harmonic mean of precision and recall. Such a balance is the inevitable outcome of the synergistic effect of the model's core components: MSFA preserves the integrity of multiscale features, SGDP provides semantic guidance, and DDCSA enables precise feature focusing and reconstruction. Both quantitative and qualitative analyses consistently demonstrate that LGSFNet exhibits strong competitiveness. It not only achieves superior numerical performance but also produces visually cleaner, more complete, and more accurate CD maps, verifying its remarkable discriminative capability and robustness in handling the inherent complexity of RSI.

D. Ablation Studies

To further validate the effectiveness of our approach, we conduct ablation studies on two representative

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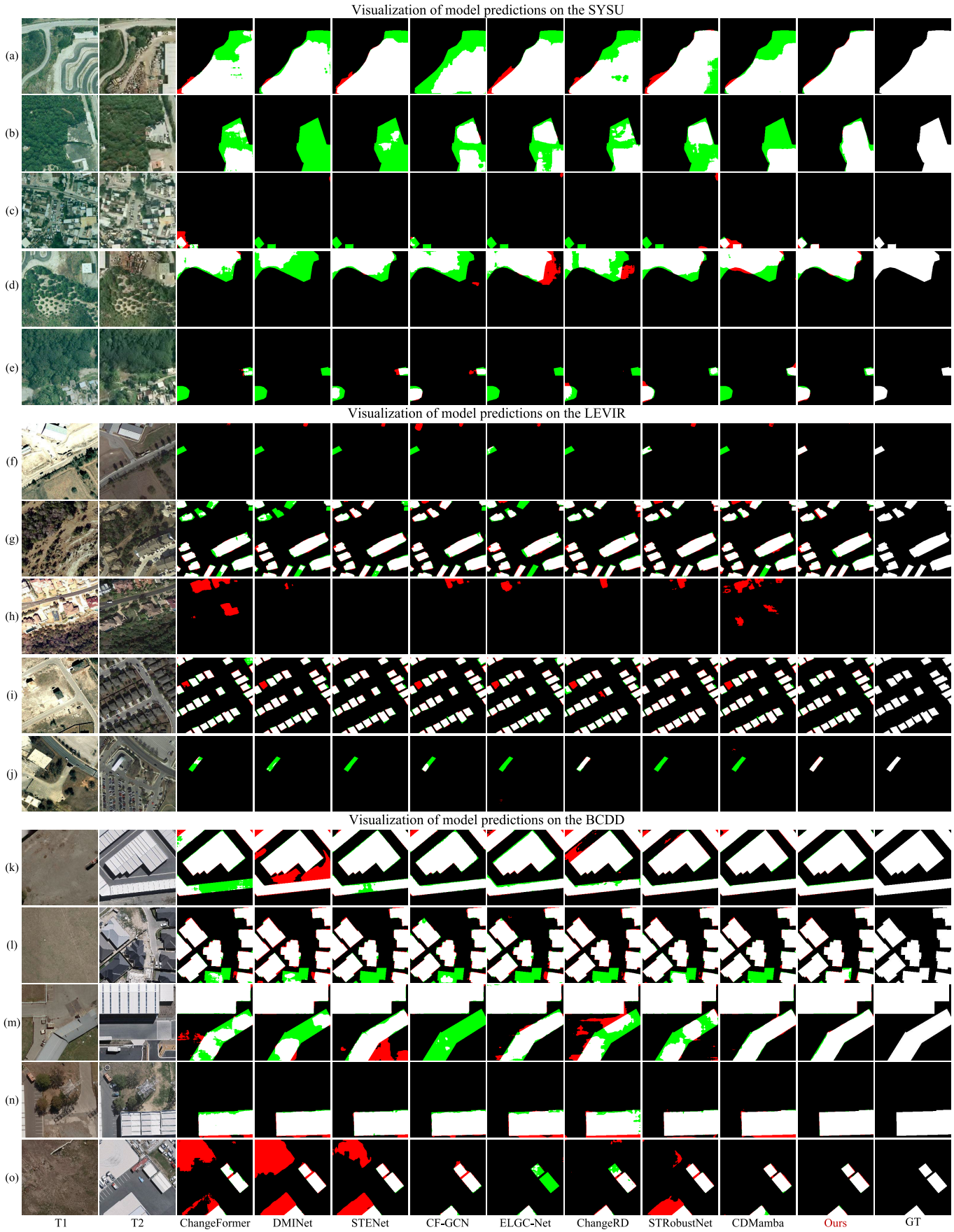


Fig. 7. Visualization of model predictions on three datasets. (a)–(o) Representative samples. GT denotes the ground truth. White: changes; black: unchanged; red: false detections; and green: missed detections.

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TABLE IV
QUANTITATIVE ABLATION RESULTS FOR MODULE NECESSITY

Model	Params.	LEVIR					SYSU				
		F1 (%)	IoU (%)	OA (%)	Rec. (%)	Pre. (%)	F1 (%)	IoU (%)	OA (%)	Rec. (%)	Pre. (%)
Baseline	2.55	90.48	82.62	99.05	88.98	92.04	81.87	69.30	91.44	81.97	81.77
(a) Without introducing textual priors											
MSFA	3.32	91.34	84.06	99.13	90.39	92.30	82.53	70.25	91.91	81.02	84.10
DDCSA	2.62	90.88	83.28	99.08	90.18	91.59	82.57	70.32	92.09	79.50	**85.89**
MSFA&DDCSA	3.39	91.57	84.45	99.15	90.62	92.54	83.19	71.22	92.25	81.31	85.17
(b) Introducing textual priors											
SGDP	28.88	91.11	83.67	99.10	90.46	91.78	81.95	69.42	91.72	79.68	84.34
SGDP&MSFA	29.65	91.63	84.56	99.16	90.48	**92.81**	82.15	69.71	91.78	80.19	84.20
SGDP&DDCSA	28.96	91.25	83.92	99.12	90.53	91.99	83.63	71.86	92.24	**84.00**	83.26
SGDP&MSFA&DDCSA	29.73	**91.77**	**84.79**	**99.17**	**91.13**	92.42	**83.79**	**72.11**	**92.38**	83.52	84.07
(c) Using the original CLIP image encoder											
Coupled CLIP Image-Text Encoder	60.94	91.56	84.43	99.15	90.48	92.66	82.88	70.77	91.96	82.54	83.23

TABLE V
QUANTITATIVE COMPARISON OF THE EFFECTS OF DIFFERENT CONFIGURATIONS ON THE PROPOSED METHOD

Type	LEVIR					SYSU				
	F1 (%)	IoU (%)	OA (%)	Rec. (%)	Pre. (%)	F1 (%)	IoU (%)	OA (%)	Rec. (%)	Pre. (%)
(a) DDCSA: Sparsity Degree Selection Analysis										
{3/10, 5/10, 7/10, 9/10}	91.36	84.10	99.14	89.74	**93.05**	83.10	71.09	92.18	81.54	84.73
{1/2, 2/3, 3/4, 4/5}	**91.77**	**84.79**	**99.17**	**91.13**	92.42	**83.79**	**72.11**	**92.38**	**83.52**	84.07
{6/10, 7/10, 8/10, 9/10}	91.47	84.28	99.14	90.39	92.57	82.97	70.89	92.16	80.94	**85.10**
(b) DDCSA: High-Frequency Subband Ablation										
HH	**91.77**	**84.79**	**99.17**	**91.13**	92.42	**83.79**	**72.11**	**92.38**	83.52	84.07
HL	91.52	84.36	99.15	90.09	**92.99**	83.67	71.93	92.30	**83.66**	83.68
LH	91.61	84.51	99.16	90.41	92.83	82.84	70.70	92.20	79.84	**86.07**
HL & LH	91.54	84.40	99.15	90.47	92.63	83.21	71.24	92.29	80.97	85.57
HH & HL	91.73	84.73	99.16	91.15	92.33	82.88	70.77	92.05	81.66	84.14
HH & LH	91.41	84.18	99.14	90.31	92.54	82.76	70.59	92.11	80.32	85.34
HL & LH & HH	91.51	84.35	99.14	90.67	92.36	83.13	71.13	92.06	82.96	83.30
(c) SGDP: Bi-Temporal Difference Modeling										
Subtraction	91.66	84.61	99.16	90.64	92.71	83.18	71.20	92.25	81.26	85.19
(d) Prompt: Different Semantic Prompt Templates										
Template 1	91.65	84.58	99.16	90.90	92.40	83.15	71.16	92.06	83.11	83.19
Template 2	**91.77**	**84.79**	**99.17**	**91.13**	92.42	**83.79**	**72.11**	**92.38**	**83.52**	84.07
Template 3	91.47	84.29	99.14	90.10	**92.89**	82.74	70.56	92.03	80.97	**84.59**
(e) Loss Functions										
BCE	91.08	83.63	99.11	89.67	**92.55**	82.35	70.00	91.79	81.23	83.51
Dice	91.06	83.59	99.10	90.12	92.02	82.58	70.33	91.91	81.35	83.85
BCE+Dice	**91.77**	**84.79**	**99.17**	**91.13**	92.42	**83.79**	**72.11**	**92.38**	**83.52**	**84.07**

datasets: LEVIR and SYSU. Results are summarized in Tables IV and V and Figs. 8 and 9.

Effectiveness of Text Priors: To validate the role of text priors in enhancing feature discriminability, we perform an ablation on SGDP to control the incorporation of text priors. As shown in Table IV, removing SGDP causes the $F1$ and IoU scores to drop by 0.20% and 0.34% on LEVIR, and by 0.60% and 0.89% on SYSU, respectively. These results indicate that text priors improve the model's competitiveness in complex scenes. As depicted in Fig. 8(b), removing text priors reduces the model's ability to distinguish change features, leading to false negatives, which visually supports their necessity. To validate the model's sensitivity to prompt templates, we conducted two additional sets of experiments [see Table V(d)].

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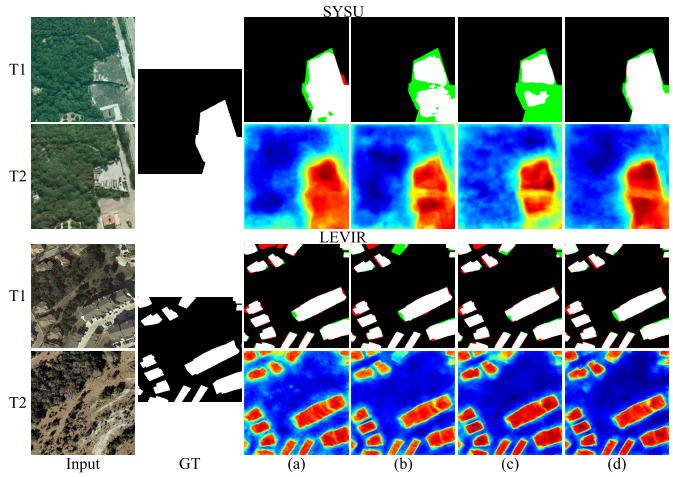


Fig. 8. Ablation visualization comparison. (a) Without MSFA. (b) Without SGDP. (c) Without DDCSA. (d) LGSFNet (ours).

The results exhibit an "inverted U-shape" trend: our standard Template 2 ("this image contains [Class$_K$]") outperforms both the minimal class-only Template 1 ("[Class$_K$]") and the complex descriptive Template 3 ("a remote sensing image showing changes related to [Class$_K$]"). We attribute the performance drop in Template 1 to the distribution gap, as CLIP favors sentence-level context over isolated labels [28]. Conversely, for Template 3, the introduction of verbose modifiers (e.g., "showing changes related to...") leads to semantic attention dilution. During text encoding, these nonsubstantive words and abstract terms occupy a portion of the feature weights, diminishing the dominance of the core category token within the feature vector. In contrast, our Template 2 strikes a balance: it provides the necessary declarative context to align with CLIP's pretraining paradigm while generating a pure, highly directive category embedding, enabling the SGDP module to localize target objects more precisely. On this basis, we also analyzed the model's sensitivity to the number of semantic categories K (number of text nodes). As illustrated in Fig. 9(a) and (b), we evaluate $K \in \{3, 4, 6, 9, 12\}$. On both the LEVIR and SYSU datasets, the model performance exhibits an initial upward trend as K increases, reaching its peak at $K = 9$. This indicates that a moderate number of semantic nodes in $\mathcal{T}$ provides a sufficiently rich categorical prior to guide the network in localizing change-related objects. However, when K exceeds 9, the performance slightly declines, likely because the inclusion of low-confidence background categories introduces semantic redundancy and noise. Moreover, in Table V(c), we explore replacing the concatenation operation in SGDP with absolute difference. This modification results in a performance drop—particularly on SYSU, where $F1$ and IoU decline by 0.61% and 0.91%. This suggests that concatenation better preserves semantic detail in text priors, improving CD. While the CLIP text encoder introduces additional parameters, its benefits in complex scenarios justify the cost.

Effectiveness of MSFA: MSFA is designed to enhance the representation capacity of lightweight backbones through progressive feature interaction. As shown in Table IV(a), when text priors are not introduced, the ablation of MSFA causes the model's $F1$ and IoU to drop by 0.69% and 1.17% on LEVIR, and by 0.62% and 0.90% on SYSU. As shown in Table IV(b), with text priors introduced, the ablation of MSFA results in a performance decrease of 0.52% in $F1$ and 0.87% in IoU on LEVIR, and 0.16% and 0.25% on SYSU, respectively. Moreover, the introduction of MSFA only adds 0.77-M parameters, demonstrating its cost-effectiveness. In addition, as illustrated in the heatmap of Fig. 8(a), the removal of MSFA increases the risk of false CD, visually validating the necessity and effectiveness of MSFA.

Effectiveness of DDCSA: DDCSA simulates the adaptive focusing mechanism of the human visual system by stacking multiple attention ranges, enabling the model to adapt to diverse change patterns. As shown in Table IV, DDCSA consistently improves performance on the SYSU, regardless of the presence of text priors. As illustrated in Fig. 8(c), removing DDCSA weakens the model's ability to discriminate changing regions, resulting in a larger number of missed detections. To determine the optimal sparsity level, we treat the sparsity ratio k as a set of key hyperparameters and conduct systematic optimization. We evaluate multiple configurations, including sparse settings (e.g., $\{3/10, 5/10, 7/10, 9/10\}$), moderate settings (e.g., $\{1/2, 2/3, 3/4, 4/5\}$), and dense settings (e.g., $\{6/10, 7/10, 8/10, 9/10\}$). As shown in Table V(a), the model achieves the best overall performance when adopting the moderate sparsity combination $\{1/2, 2/3, 3/4, 4/5\}$. We argue that this configuration achieves a favorable balance between preserving sufficient global context (preventing over-sparsity) and focusing on key local changes (preventing over-density). Therefore, we finalize the DDCSA design based on the performance observed in this hyperparameter study. On this basis, we visualize the feature processing at this sparse ratio in Fig. 10. From left to right, as the sparsity ratio increases, more details of channel interactions are included in the computation. The contrast between the change region and the background is gradually widened. This further validates the effectiveness of our design. In addition, we investigate the influence of different high-frequency sub-band combinations (HL, LH, and HH) on the final performance. As shown in Table V(b), compared with the conventional design that employs all high-frequency sub-bands simultaneously, the simplified configuration retaining only the HH sub-band achieves superior performance within our framework. We believe this configuration strikes a better balance between computational complexity and feature representation capability. Therefore, motivated by engineering considerations to simplify the model structure while pursuing higher performance, we ultimately adopt the HH-only configuration as our final design choice.

Discussion of Loss Functions: The ablation results in Table V(e) demonstrate the effectiveness of our integrated objective function combining BCE and Dice losses. This joint optimization strategy demonstrates a stronger capability in mitigating data imbalance compared to a single supervised approach. Specifically, BCE reinforces fine-grained pixel discrimination, while Dice focuses on global structural alignment, which is a significant advantage in scenarios with a sparse distribution of changing pixels.

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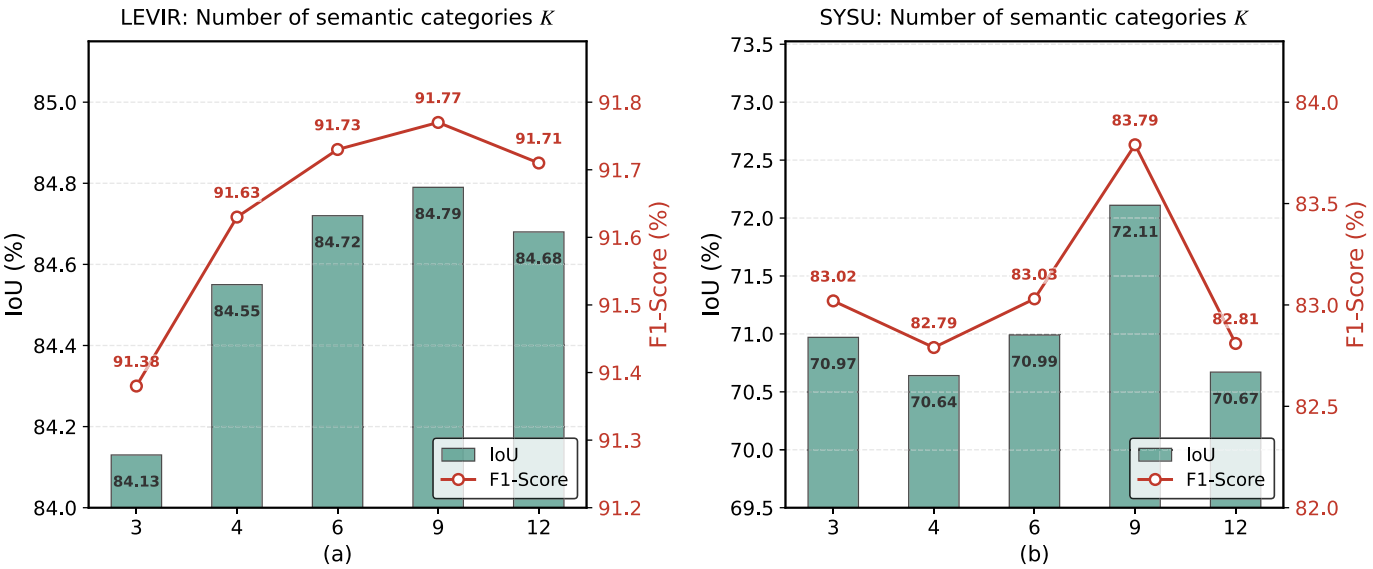


Fig. 9. Ablation study on semantic richness. (a) and (b) Ablation results on semantic richness for the LEVIR and SYSU datasets, respectively.

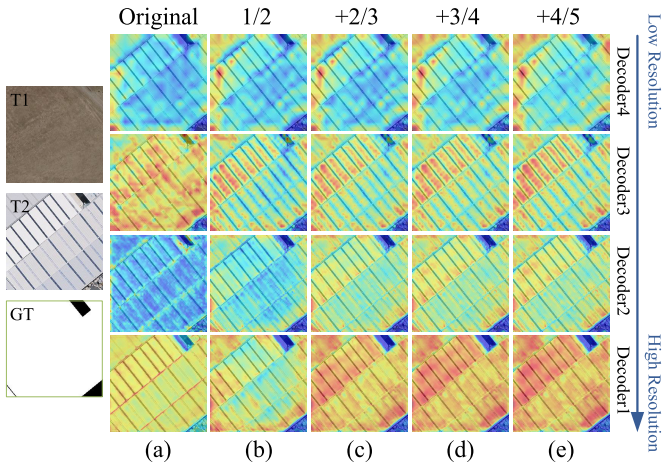


Fig. 10. Visualization of progressive aggregated features in DDCSA modules across different decoder levels. Rows (top to bottom) correspond to decoder stages from deep to shallow layers. From left to right, the sequence illustrates the progressive refinement of spatial discriminability for ground objects, achieved by increasing the complexity of channel interactions based on sparsity ratios of $\{1/2, 2/3, 3/4, 4/5\}$. (a)-(e) Original features and the progressively aggregated results obtained by sequentially applying sparsity configurations of 1/2, then adding 2/3, 3/4, and 4/5, respectively.

V. Discussion

Although the proposed method demonstrates clear advantages in overall performance, several representative limitations can still be observed from the red-green visualization results shown in Fig. 7. First, LGSFNet continues to face challenges in accurately delineating complex and fine-grained change boundaries. As illustrated by typical samples from the three datasets in Fig. 7, the model is able to localize the main change regions with relatively high accuracy. However, partial false detections and incomplete coverage are still observed along the boundaries of building and land-use changes. This phenomenon is particularly evident in the SYSU dataset, which contains a large number of irregular boundaries, where some thin or curved change edges appear blurred or discontinuous in the predicted results. We consider this limitation to mainly stem from the lightweight visual backbone, as well as the fact that DDCSA is designed to enhance semantic-aware channel interactions rather than explicitly modeling fine-grained boundary geometry. Although the introduced frequency-domain information alleviates this issue to some extent, precise boundary reconstruction remains challenging. In addition, as validated by the ablation experiments, the textual prompts used in LGSFNet are intentionally designed to be concise and label-oriented (e.g., "this image contains [Class$_K$]"), focusing on encoding category-level semantics. While more descriptive prompts may provide richer semantic cues, they often introduce semantic dilution or noise, which can adversely affect pixel-level change localization. This design reflects a deliberate tradeoff between semantic expressiveness and localization stability. In recent years, large-scale vision–language models have increasingly adopted richer textual descriptions to enhance semantic modeling, a trend that suggests the potential value of more detailed prompt information. However, how to reliably incorporate such detailed textual representations into pixel-level CD, while maintaining localization accuracy and robustness, remains an open and challenging problem. In summary, simultaneously enhancing fine-grained boundary delineation and semantic representation capability, while maintaining model robustness and computational efficiency, remains a key challenge commonly faced by semantic-driven image CD methods and warrants further investigation in future work.

VI. Conclusion

In this article, we propose a detachable text prior introduction method. This provides a new perspective on how to use text prior to RSCD tasks. Different from existing frameworks, we combine the CLIP text encoder with a lightweight visual backbone, which significantly reduces the model complexity while improving feature discriminability. In addition, MSFA is proposed to make up for the feature representation ability

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of the lightweight backbone. On this basis, we design a lightweight DDCSA. It combines the advantages of both spatial and frequency domains. This enables effective modeling of diverse change patterns at low cost and helps suppress interference from background noise. In summary, we develop a novel CD architecture, called LGSFNet. Experiments on three mainstream CD datasets show that LGSFNet has strong competitiveness. However, LGSFNet still has some limitations. For example, its ability to characterize complex change boundaries requires further improvement. Future work will focus on enhancing edge perception capabilities to better handle intricate structural changes.

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